

Adam Sultan

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

B.S. in Information Systems & Human-Computer Interaction; GPA: 3.56/4.0

Expected May 2027

Relevant Coursework: Data Structures & Algorithms, Computer Systems, Software Engineering Foundations, Software Construction, Application Design & Development, Intro to Machine Learning

EXPERIENCE

The Silicon Project

New York, NY

Board Director & Lead Developer

Jun 2022 – Present

- Co-architected the organization's platform using **Next.js** and **Supabase Auth/Postgres**; built an internal employee portal and a **Stripe-powered** e-commerce store.
- Developed secure backend systems to track donor data and inventory logistics, enabling the processing of **\$422K+** in donations and **12,000+ lbs** of e-waste.
- Established engineering standards including **GitHub** branching strategies, PR review workflows, and reusable component libraries to scale volunteer contributions.

ENYX Therapeutics LLC

New York, NY

Software Developer Intern

May 2025 – Aug 2025

- Spearheaded end-to-end development of the company website using **React.js** and **Tailwind CSS**, creating a componentized UI with robust routing and form handling.
- Collaborated with leadership to translate **Figma** designs into responsive layouts; optimized accessibility and shipped prioritized pages ahead of schedule.

Intelliminds

New York, NY

Founding Software Engineer

Aug 2023 – Feb 2025

- Engineered a custom CRM web application using **Next.js**, **TypeScript**, and **Supabase** to automate client-tutor scheduling, reducing manual coordination time by **50%**.
- Implemented SEO optimization and analytics tracking for the public-facing site, driving **1,000+** unique visits and increasing client acquisition.
- Led a team of developers through **Agile sprints**, managing the full SDLC from feature definition to deployment.

PROJECTS

NodeBB Open-Source Contributor | *JavaScript/Node.js, Redis, GitHub Actions, Stryker*

Sep 2025 – Nov 2025

- Developed a "Post Anonymously" plugin in a team fork; implemented server/client hooks, admin settings UI, and permission checks (16 PRs merged).
- Configured **CI/CD pipelines** with GitHub Actions and introduced Stryker mutation testing to harden edge cases and ensure upstream API compatibility.

Concurrent HTTP Proxy Server | *C, Sockets, Pthreads, HTTP, Synchronization*

Nov 2025

- Built a concurrent HTTP/1.0 proxy in C using POSIX sockets and **pthread**s; handled multiple clients with robust request parsing/forwarding and **Rio** buffered I/O.
- Implemented a thread-per-connection model with detached threads and defensive error handling to prevent resource leaks and improve reliability under concurrent load.

Dynamic Memory Allocator | *C, Git, Valgrind, GDB*

Oct - Nov 2025

- Built a 64-bit memory allocator in C (**malloc/free/realloc/calloc**) with **16-byte alignment** and heap growth via **mem_sbrk**.
- Reduced fragmentation with block **splitting/coalescing**; debugged and verified heap invariants using a **heap checker**, **mdriver-dbg**, **GDB**, and **Valgrind**.

TECHNICAL SKILLS

Languages: Java, C, Python, TypeScript, JavaScript, SQL (PostgreSQL), HTML/CSS, R, Ruby

Frameworks: React, Next.js, Node.js, Tailwind CSS, Ruby on Rails

Developer Tools: Git/GitHub, Docker, Linux/Unix, Make, GDB, Valgrind, Figma, Vercel, Supabase

Concepts: Systems Programming, REST APIs, Agile/Scrum, CI/CD, MVC Architecture, Database Design